Marginalisation & Type-II Maximum Likelihood

Realising the Power of Bayesian Inference

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2026-10-02

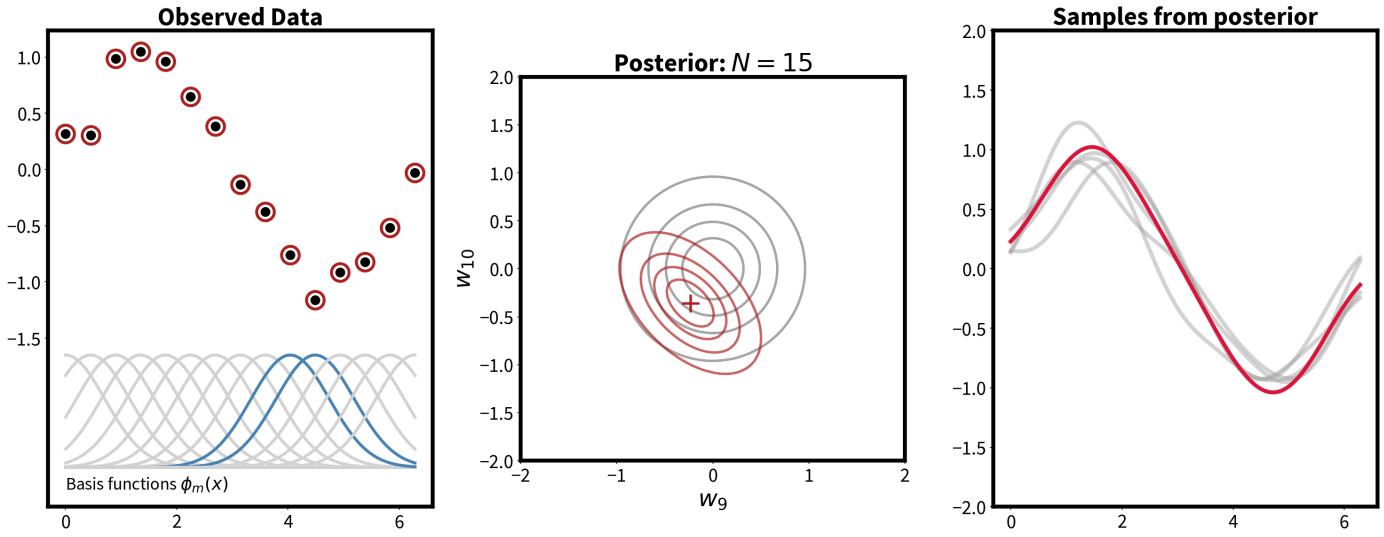
Slide 1 In this Episode ...

- Maximum *a posteriori* (MAP) estimation — brief review
- Marginalisation: the Bayesian way to make predictions
- Type-II Maximum Likelihood: approximate inference in the Bayesian linear regression model
- The marginal likelihood and “Occam’s razor”
- Inferring both hyper-parameter α and noise level σ^2 from the training data
- Prediction revisited

Recap

Slide 2 Recap: Bayesian MAP Estimation

The posterior distribution, and posterior mean (MAP) predictor (right), after observing the 15 data points:



Note: We'd assumed we knew the generative noise variance σ^2 and we arbitrarily chose a "reasonable value" of α

Slide 3 MAP -vs- Marginalisation

- In linear regression, we saw that the maximum *a posteriori* (MAP) estimate is equivalent to the penalised least-squares (PLS) estimate: $\boldsymbol{\mu} = \widehat{\mathbf{w}_{PLS}}$
- It might be thought that the Bayesian framework is simply a probabilistic re-interpretation of classical methods
- This is not the case!

- The distinguishing element of Bayesian methods is marginalisation, the integrating out of all nuisance variables
- As we will now see, this is a powerful component of the Bayesian framework

Key Point Maximum *a posteriori* (MAP) parameter estimation is **not** "Bayesian" !!

Slide 4 Recap: The Bayesian Linear Regression Framework

- **Likelihood model** (same as orthodox):

$$p\left(\mathbf{y}|\mathbf{w},\sigma^2\right) = \left(2\pi\sigma^2\right)^{-N/2} \prod_{n=1}^{N} \exp\left\{ -\frac{\left[y_n - \widehat{f}\left(x_n;\mathbf{w}\right)\right]^2}{2\sigma^2} \right\}$$

- **Prior** (incorporating precision hyper-parameter α):

$$p\left(\mathbf{w}|\alpha\right) = \prod_{m=1}^{M} \left(\frac{\alpha}{2\pi}\right)^{1/2} \exp\left\{ -\frac{\alpha}{2} w_m^2 \right\}$$

- **Posterior** (from Bayes' rule): $p\left(\mathbf{w}|\mathbf{y},\alpha,\sigma^2\right) = \mathcal{N}\left(\boldsymbol{\mu},\boldsymbol{\Sigma}\right)$ with

$$\boldsymbol{\mu} = \left(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi} + \sigma^2\alpha\mathbf{I}\right)^{-1}\boldsymbol{\Phi}^{\mathrm{T}}\mathbf{y} \Longrightarrow \text{equivalent to } \widehat{\mathbf{w}}_{PLS}$$

$$\boldsymbol{\Sigma} = \sigma^2\left(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi} + \sigma^2\alpha\mathbf{I}\right)^{-1}$$

Being “Strictly Bayesian”

Slide 5 Strictly Bayesian: the Full Predictive Framework (1)

In general, for any model, if we wish to predict y_* (at x_*) given some training data **y**, what we really, really, want is:

$$p\left(y_*|x_*, \mathbf{y}\right)$$

$$p\left(y_*|\mathbf{y}\right)$$

Slide 6 Strictly Bayesian: the Full Predictive Framework (2)

- So far, we’ve only placed a prior over the weights **w** — to be strictly Bayesian we should define:
  - $p\left(\alpha\right)$ — a hyper-prior over the hyper-parameter, and
  - $p\left(\sigma^2\right)$ — a prior over the noise variance
- Then we need to compute the full posterior over all nuisance variables: $p\left(\mathbf{w}, \alpha, \sigma^2|\mathbf{y}\right)$
- ... and use it to make the desired predictions:

$$p\left(y_*|\mathbf{y}\right) = \int p\left(y_*|\mathbf{w}, \alpha, \sigma^2\right) p\left(\mathbf{w}, \alpha, \sigma^2|\mathbf{y}\right) d\mathbf{w}\, d\alpha\, d\sigma^2$$

- This prediction incorporates our uncertainty over all unknown parameters by averaging over their likely values having seen the data **y**

Slide 7 Strictly Bayesian: the Full Predictive Framework (3)

- So we need the joint posterior over *all* unknowns
- From Bayes' rule, this is:

$$p\left(\mathbf{w}, \alpha, \sigma^2 | \mathbf{y}\right) = \frac{p\left(\mathbf{y}|\mathbf{w}, \sigma^2\right) p\left(\mathbf{w}|\alpha\right) p\left(\alpha\right) p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)}$$

- The highlighted normalising factor is the "fully marginalised" likelihood:

$$p\left(\mathbf{y}\right) = \int p\left(\mathbf{y}|\mathbf{w}, \sigma^2\right) p\left(\mathbf{w}|\alpha\right) p\left(\alpha\right) p\left(\sigma^2\right) \, d\mathbf{w} \, d\alpha \, d\sigma^2$$

- It is nearly always analytically intractable!
- Nevertheless, as we'll soon see, $p\left(\mathbf{y}\right)$ is an extremely useful quantity, even when approximated

Slide 8 Making Predictions: Summary of Approaches

Summary of "learning" options, in increasing order of "morality":

Framework	Learned Quantity	Prediction	Lecture(s)
Orthodox	$\widehat{\mathbf{w}_{PLS}}$	$\widehat{f}\left(x_*; \widehat{\mathbf{w}_{PLS}}\right)$	5
MAP	$p\left(\mathbf{w}|\mathbf{y}, \alpha, \sigma^2\right)$	$p\left(y_*|\widehat{\mathbf{w}_{MAP}}, \sigma^2\right)$	5
Basic Bayesian	$p\left(\mathbf{w}|\mathbf{y}, \alpha, \sigma^2\right)$	$p\left(y_*|\mathbf{y}, \alpha, \sigma^2\right)$	5
Type-II Bayesian	$p\left(\mathbf{w}|\mathbf{y}, \widehat{\alpha}_{MP}, \widehat{\sigma}^2_{MP}\right)$, $\widehat{\alpha}_{MP}, \widehat{\sigma}^2_{MP}$	$p\left(y_*|\mathbf{y}, \widehat{\alpha}_{MP}, \widehat{\sigma}^2_{MP}\right)$	

Framework	Learned Quantity	Prediction	Lecture(s)
Strict Bayesian	$p\left(\mathbf{w}, \alpha, \sigma^2 \middle\| \mathbf{y}\right)$	$p\left(y_* \middle\| \mathbf{y}\right)$	7–10

Colour indicates how parameter uncertainty is handled:

Fully modelled
Approximately captured
Ignored!

Slide 9 **The Analytic Roadblock**

“Strict Bayesian” inference:

- In our model would be:

$$p\left(y_* \middle\| \mathbf{y}\right) = \int p\left(y_* \middle\| \mathbf{w}, \sigma^2\right)\, p\left(\mathbf{w}, \alpha, \sigma^2 \middle\| \mathbf{y}\right)\, d\mathbf{w}\, d\alpha\, d\sigma^2$$

- We can’t compute either $p\left(\mathbf{w}, \alpha, \sigma^2 \middle\| \mathbf{y}\right)$ or $p\left(y_* \middle\| \mathbf{y}\right)$ analytically

Alternative approach:

1. Perform analytically computable integrations (in some extremely specific cases, you may be able to marginalise one or two hyperparameters too), then
2. Approximate remaining terms, or entire integrals, perhaps by:
    - Type-II maximum likelihood

- Laplace's method
- Monte-Carlo sampling
- Variational techniques

Type-II Maximum Likelihood

Slide 10 Type-II Maximum Likelihood (1)

- The desired posterior, $p\left(\mathbf{w}, \alpha, \sigma^2|\mathbf{y}\right)$, can be written as (product rule):
$$p\left(\mathbf{w}, \alpha, \sigma^2|\mathbf{y}\right) \equiv p\left(\mathbf{w}|\mathbf{y}, \alpha, \sigma^2\right)\ p\left(\alpha, \sigma^2|\mathbf{y}\right)$$
- First term is the weight posterior: $p\left(\mathbf{w}|\mathbf{y}, \alpha, \sigma^2\right) = N\left(\mathbf{w}|\boldsymbol{\mu}, \boldsymbol{\Sigma}\right)$
- Second term $p\left(\alpha, \sigma^2|\mathbf{y}\right)$ is intractable:
  - we will approximate it (very coarsely!) with a δ-function at its mode
- We find most probable values $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ which maximise:
$$p\left(\alpha, \sigma^2|\mathbf{y}\right) = \frac{p\left(\mathbf{y}|\alpha, \sigma^2\right)\ p\left(\alpha\right)\ p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)}$$

Slide 11 Type-II Maximum Likelihood (2)

- We want to find the maximum of the posterior:
$$p\left(\alpha, \sigma^2|\mathbf{y}\right) = \frac{p\left(\mathbf{y}|\alpha, \sigma^2\right)\ p\left(\alpha\right)\ p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)}$$

- If we assume flat uninformative priors over $\log \alpha$ and $\log \sigma$, then we equivalently maximise $p\left(\mathbf{y}|\alpha, \sigma^2\right)$
- $p\left(\mathbf{y}|\alpha, \sigma^2\right)$ is the marginal likelihood (or "evidence")
- This procedure is known as Type-II maximum likelihood
- Also, in certain contexts, the evidence procedure

Slide 12 The Marginal Likelihood (1)

- To find $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ we maximise the marginal likelihood
- This is given by:

$$p\left(\mathbf{y}|\alpha, \sigma^2\right) = \int p\left(\mathbf{y}|\mathbf{w}, \sigma^2\right) p\left(\mathbf{w}|\alpha\right) d\mathbf{w}$$

$$= (2\pi)^{-\frac{N}{2}} \left|\sigma^2\mathbf{I} + \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^{\mathrm{T}}\right|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\mathbf{y}^{\mathrm{T}}\left(\sigma^2\mathbf{I} + \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^{\mathrm{T}}\right)^{-1}\mathbf{y}\right\}$$

- This is a zero-mean Gaussian distribution over the single N-dimensional dataset vector $\mathbf{y}$ with $N \times N$ covariance matrix:

$$\sigma^2\mathbf{I} + \alpha^{-1}\mathbf{\Phi}\mathbf{\Phi}^{\mathrm{T}}$$

- It gives us a probability for the entire data set, conditional on values of α and σ^2

Slide 13 The Marginal Likelihood (2)

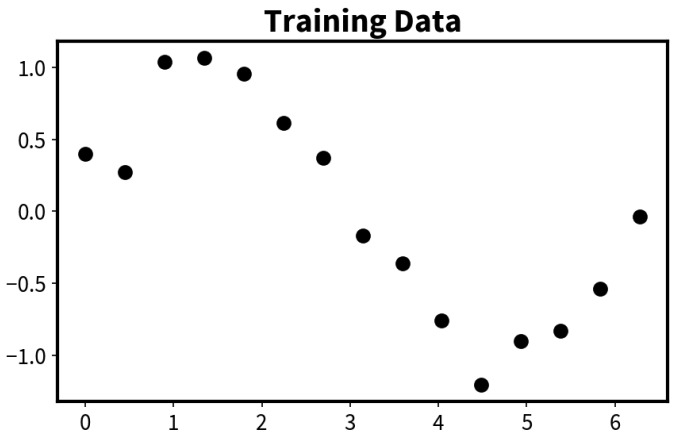
- Computing the marginal likelihood is straightforward
  - it's just a Gaussian p.d.f., although high-dimensional

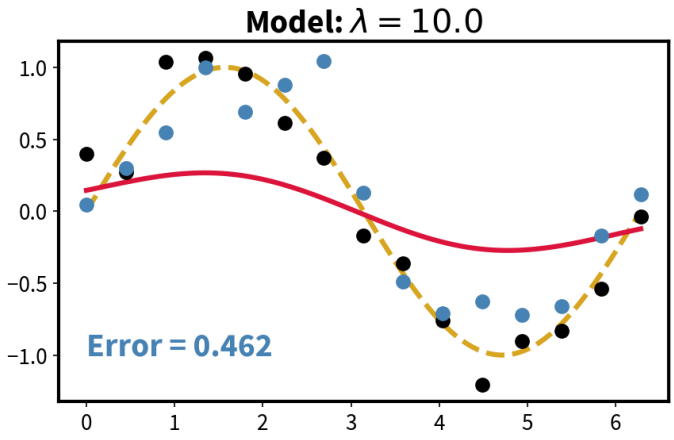
- Having computed it, we can:
  - evaluate $p\left(\mathbf{y}|\alpha,\sigma^2\right)$ over a range of values of α and σ^2
  - apply gradient methods to maximise it directly (iteratively)
- **Note:** we can use all the training data to determine $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$ — we don't need to reserve a separate validation set
  - Marginal likelihood already accounts for model complexity (Occam's Razor).
- This is clearly not "strictly Bayesian"! But it is a much more faithful, and practically useful, procedure than MAP estimation

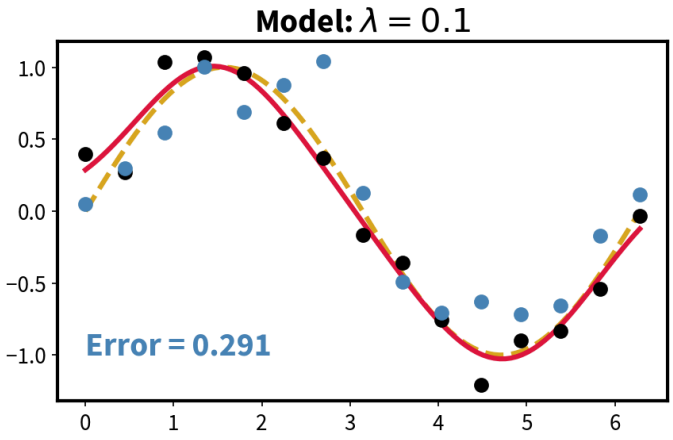
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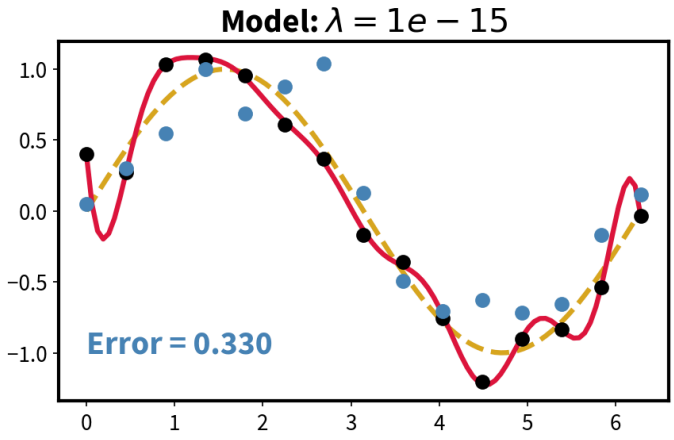
Slide 14 Recap: Impact of Regularisation Parameter λ

An appropriate value of λ in the penalised least-squares approach can be selected by holding out a separate validation set

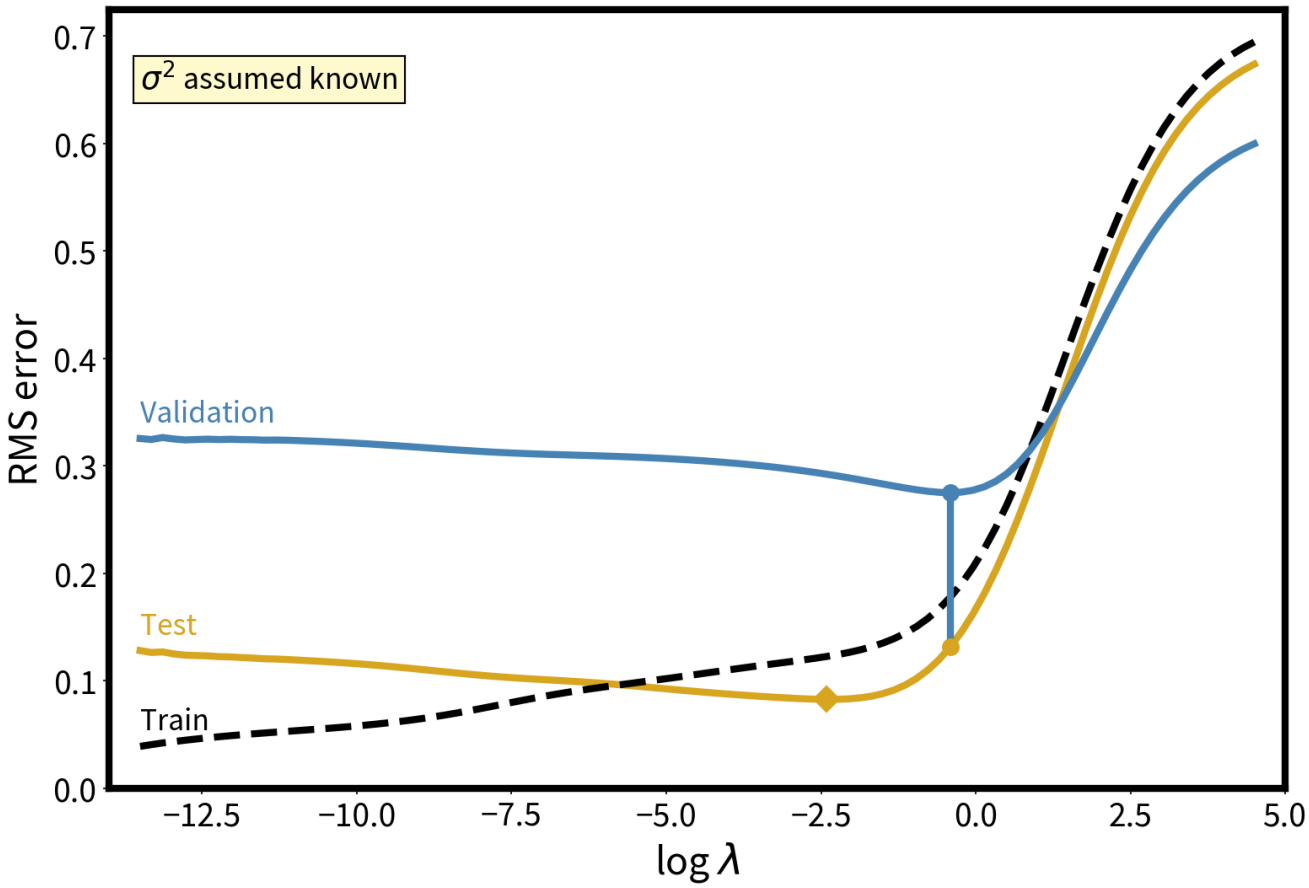

Training Data


Model: $\lambda = 10.0$
Error = 0.462

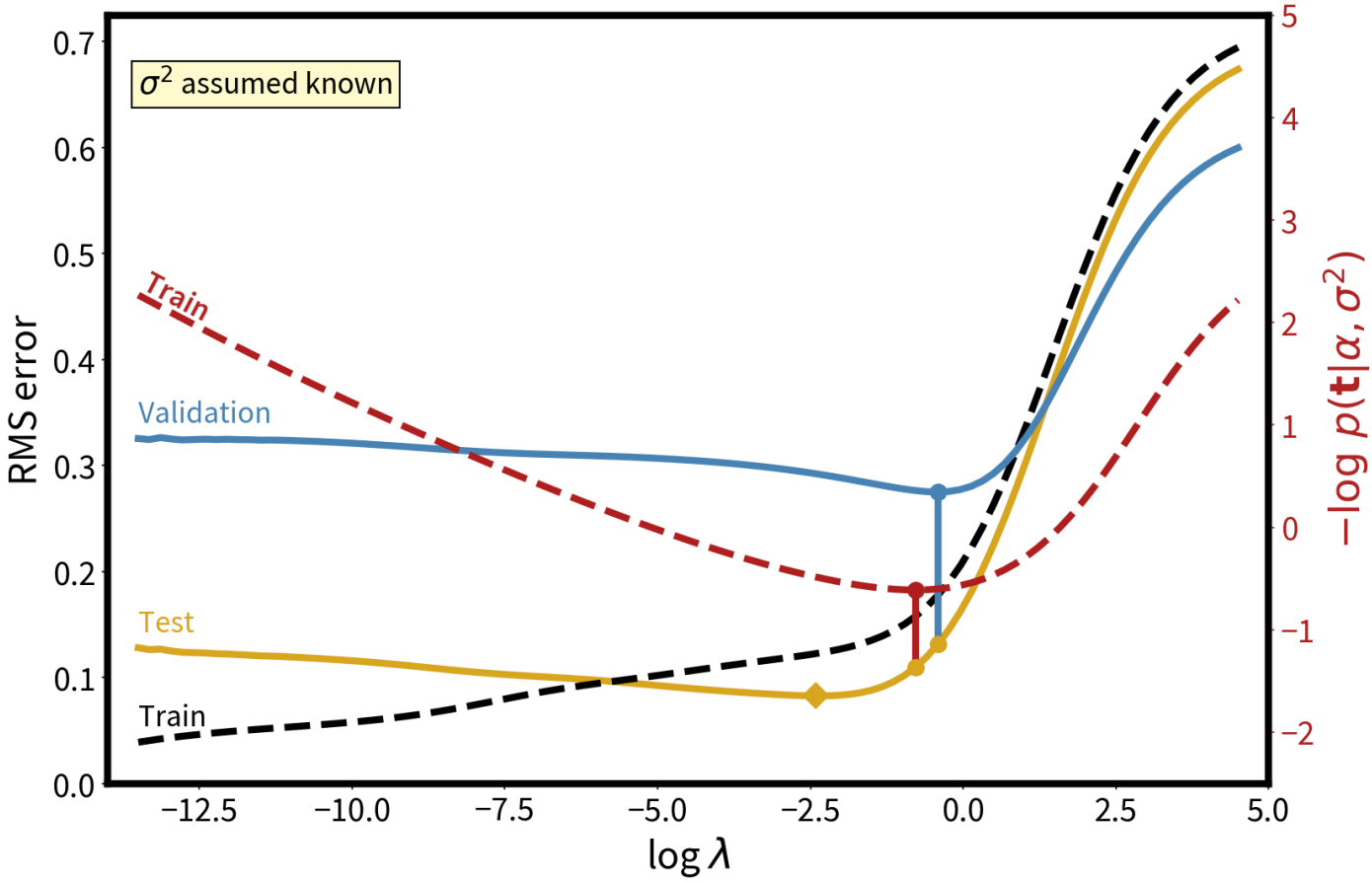

Model: $\lambda = 0.1$
Error = 0.291


Model: $\lambda = 1e-15$
Error = 0.330

Slide 15 Recap: Validation of Regularisation Parameter λ



Slide 16 Marginal Likelihood to Estimate Hyper-parameter $\widehat{\alpha}_{MP}$



Slide 17 The Bottom Line

- *Using only 15 examples and no validation data*, the Bayesian approach for setting α finds a closer model to the 'oracle':

	Orthodox	Bayesian	Oracle
Test Error	0.132	0.110	0.083

- The marginal likelihood criterion penalises models that are both too simple and too complex. This is not due to the prior.

Reminder We assumed an **uninformative** (flat) hyper-prior over $\log \alpha$, which states no preference for any complexity of model.

- This effect is often referred to as Occam's Razor (sometimes, "Ockham's")

Occam's Razor

Slide 18 Yes, There Really is Such a Place as "Occam"



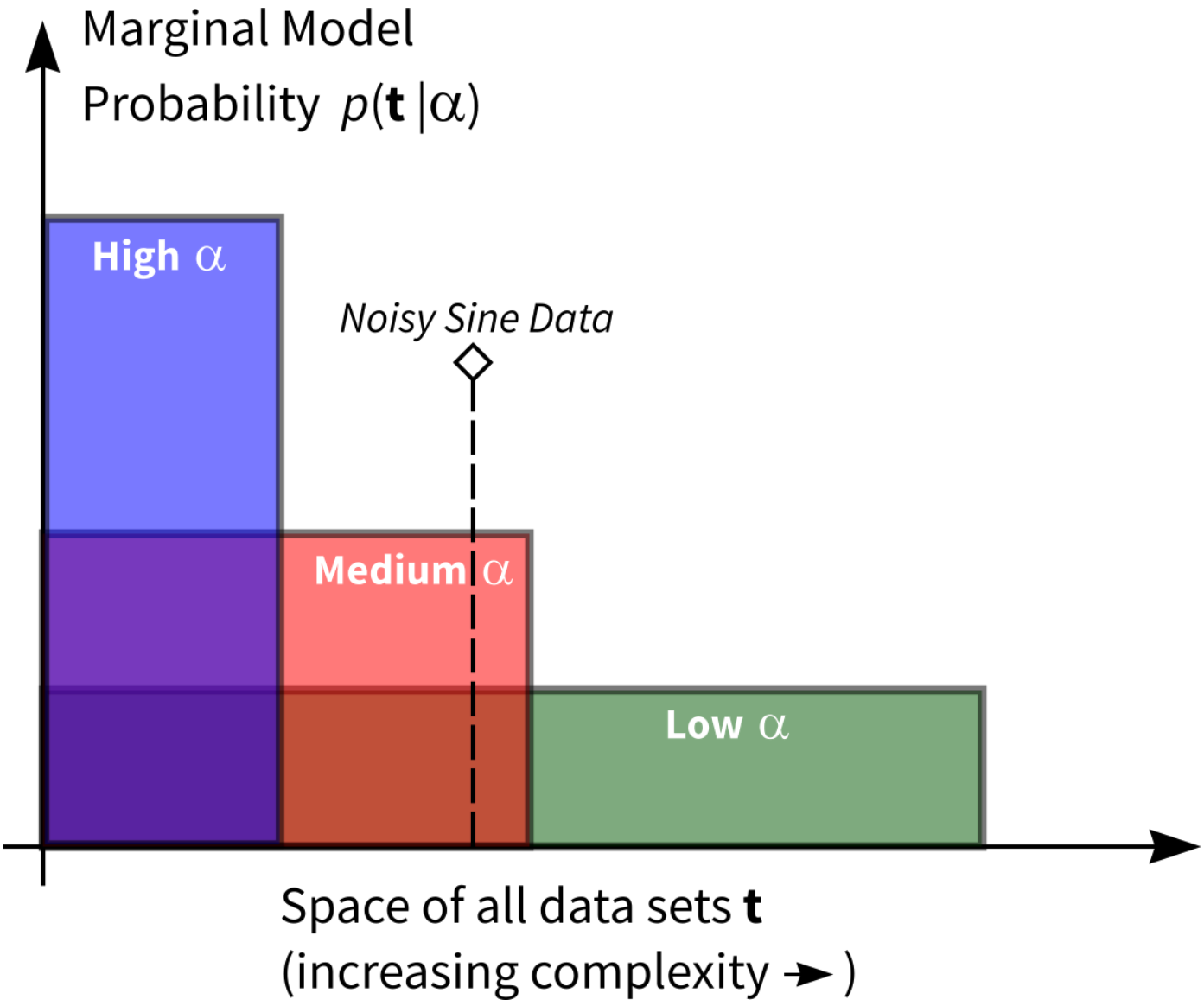
Slide 19 Occam's Razor

"Pluralitas non est ponenda sine neccesitate." *William of Occam — 14th Century*

- Literally: **"** entities should not be multiplied unnecessarily**"**
- In the context of machine learning and data science:models should be no more complex than is sufficient to explain the data
- The Bayesian procedure implicitly implements "Occam's Razor": it assigns lower probability *both* to models that are too simple *and* too complex

Key Question Why is an intermediate value of α marginally more probable?

Slide 20 Preference for the "Just Right" Model



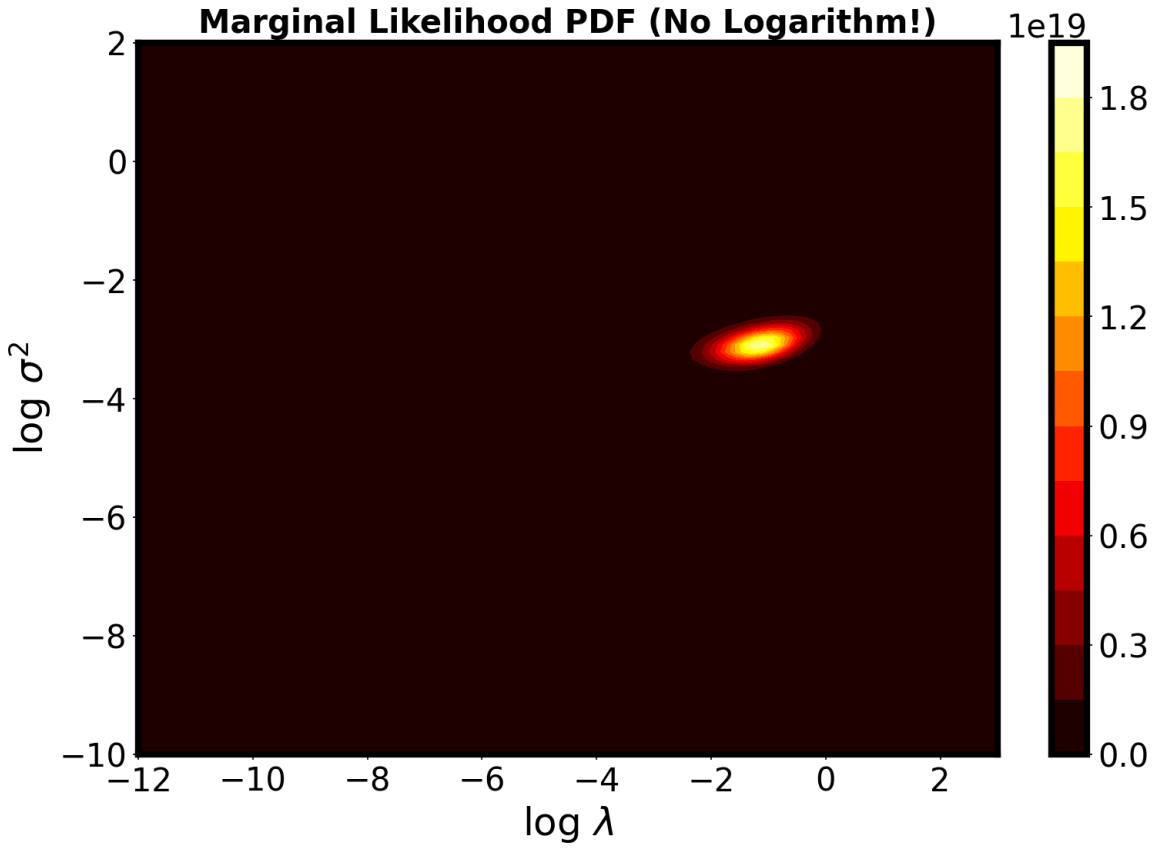
Slide 21 Next Steps: The Noise Variance

- For simplicity, the noise variance σ^2 was assumed known ($\sigma = 0.25$ in fact)
- In the (Type-II) Bayesian framework, we can concurrently optimise the marginal likelihood to find *both* $\widehat{\alpha}_{MP}$ and $\widehat{\sigma}^2_{MP}$

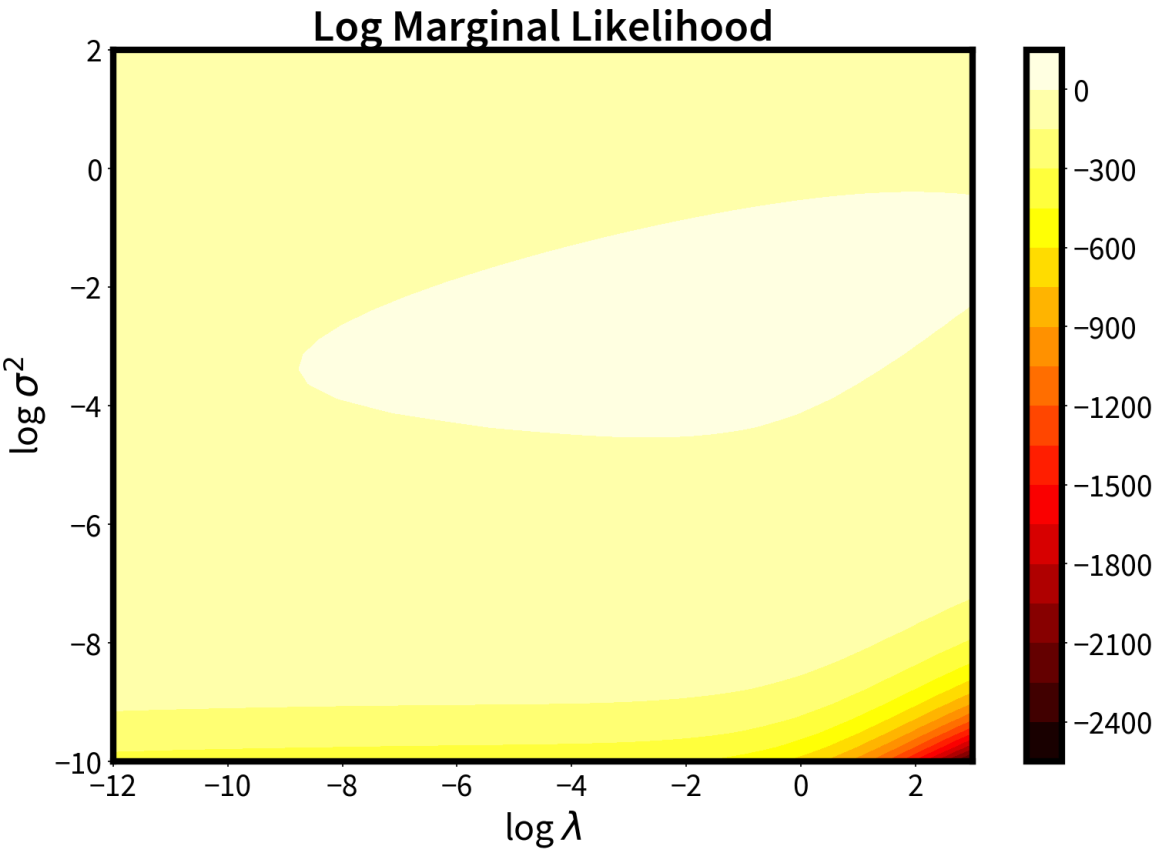
- Again, we could use gradient methods, but here we will simply compute $\log p\left(\mathbf{y}|\alpha, \sigma^2\right)$ over a range of values of α and σ^2 ...

Key point This joint optimisation would be impossible in the orthodox framework: minimising the penalised least squares measure $E_{PLS}\left(\mathbf{w}, \lambda, \sigma^2\right)$ would result in both $\lambda \to 0$ and $\sigma^2 \to 0$. The subsequent model would therefore badly over-fit.

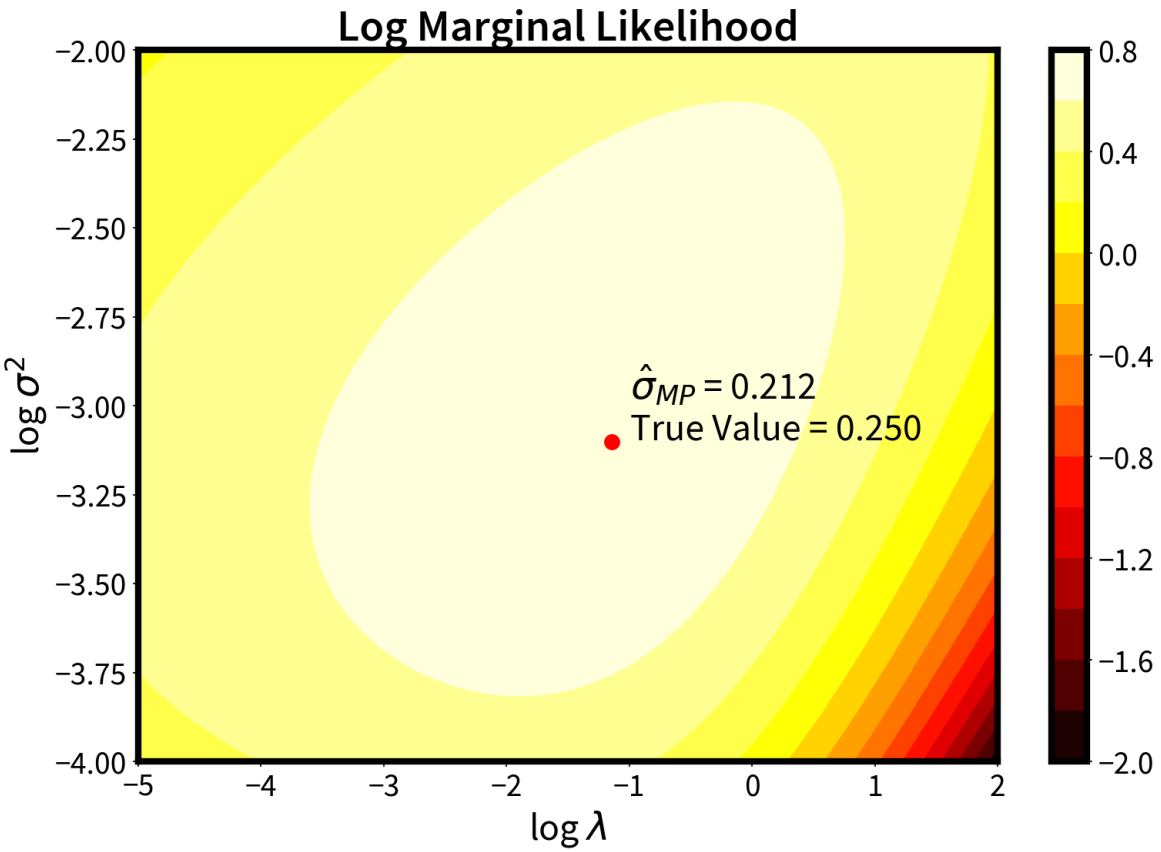
Slide 22 Concurrent Estimation of λ and Observation Noise σ^2



Slide 23 Concurrent Estimation of λ and Observation Noise σ^2



Slide 24 Concurrent Estimation (Zoomed In)



The Predictive Distribution

Slide 25 The Predictive Distribution: Type-II Approach

- The idealised predictive distribution at x_* would be:

$$p\left(y_*|\mathbf{y}\right) = \int p\left(y_*|\mathbf{w},\sigma^2\right)\, p\left(\mathbf{w},\alpha,\sigma^2|\mathbf{y}\right)\, d\mathbf{w}\, d\alpha\, d\sigma^2$$

- Instead, here we compute:

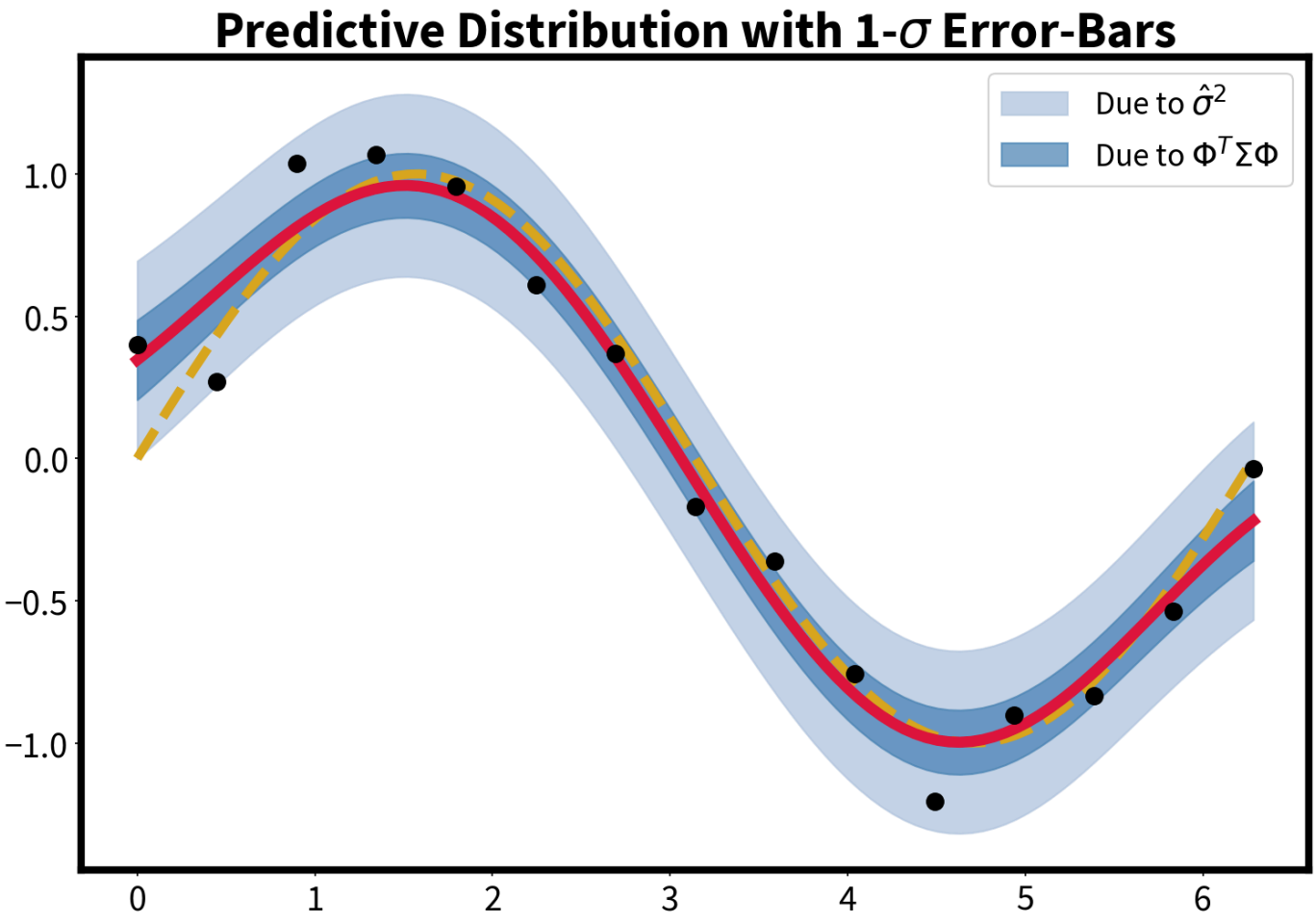
$$\begin{aligned} p\left(y_*|\mathbf{y},\widehat{\alpha}_{MP},\widehat{\sigma}^2_{MP}\right) &= \int p\left(y_*|\mathbf{w},\widehat{\sigma}^2_{MP}\right)\, p\left(\mathbf{w}|\mathbf{y},\widehat{\alpha}_{MP},\widehat{\sigma}^2_{MP}\right)\, d\mathbf{w} \\ &= \mathcal{N}\left(\boldsymbol{\phi}\left(x_*\right)^{\mathrm{T}}\boldsymbol{\mu},\sigma^2_*\right) \end{aligned}$$

- This has variance:

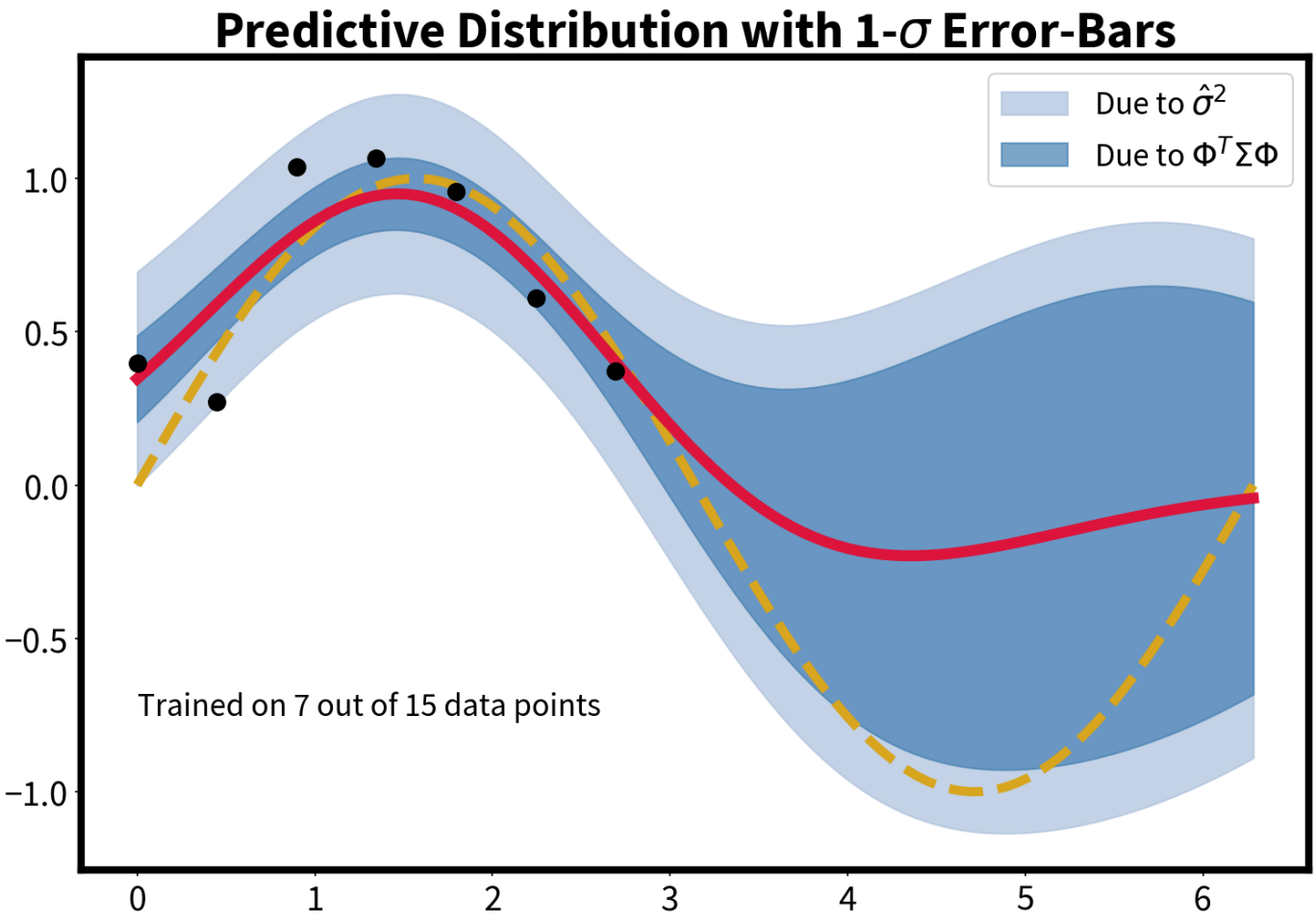
$$\sigma_*^2 = \hat{\sigma}_{MP}^2 + (x_*)^{\mathrm{T}} \quad (x_*)$$

Predictive uncertainty — Data noise — Parameter uncertainty

Slide 26 The Predictive Distribution: Type-II Approach



Slide 27 The Predictive Distribution: Type-II Approach



Next Steps

Slide 28 Next Steps: Other Unknowns and Model Selection

- Our models are also conditioned on other variables we have so far ignored: the choice of basis set ϕ and any associated parameters (for our Gaussian RBF, the width parameter r)
- Ideally, we should define priors $P(\phi)$ and $p(r)$, and integrate out those variables when making predictions

- More simply perhaps, we could use $p\left(\mathbf{y}|\phi, r\right)$ as a criterion for model selection

- For this model, it is feasible to integrate out α and σ^2 numerically to compute $p\left(\mathbf{y}|\phi, r\right)$ — we will explore that in Lecture 7.